# Pro-Active Inference

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## Abstract

## 1 Introduction

### 1.1 Related Work

## 2 Method: Pro-Active Inference

### 2.1 Finite-Population Problem Formulation and Notation

Suppose our data is comprised of pairs $\{(x_i, y_i)\}_{i=1}^N$, viewed for now as a fixed finite population. We have access to all of the covariates $x_i$, but do not have access to any of the labels $y_i$. Our estimand of interest is the finite population mean:

$$\theta := \frac{1}{N}\sum_{i=1}^{N} y_i. \tag{1}$$

Given a fixed labeling budget $n$, the problem is: how should we adaptively select which $n$ labels to query so as to construct a valid confidence interval (CI) for $\theta$ that is as narrow as possible?

At each labeling round $t = 1, \ldots, n$, we sample a label index $I_t \in [1 : N] := \mathcal{I}$ from an adaptive distribution $q_t(\cdot)$. Let $\mathcal{F}_{t-1} := \sigma\left(\{I_s, y_{I_s}\}_{s=1}^{t-1}\right)$ and let $O_{t-1} \subseteq \mathcal{I}$ be the set of label indices we have observed up to and including round $t-1$, initialized at $O_0 = \emptyset$. Because we wish to sample without replacement, at each round $t$, $q_t(\cdot)$ conditional on $\mathcal{F}_{t-1}$ is a probability distribution over $\mathcal{I} \setminus O_{t-1}$. Finally, let $\hat{f}^{(t-1)} \in m\mathcal{F}_{t-1}$ denote a potentially-black-box machine learning (ML) model intended so that $\hat{f}^{(t-1)}(x) \approx y$, though we make no assumptions on the fidelity or well-specification of $\hat{f}^{(t-1)}$. The superscript $(t-1)$ indicates that after observing each newly-queried label, we may update our ML model accordingly. We will present examples of empirically high-performing combinations of $q_t(\cdot)$ and $\hat{f}^{(t-1)}$ in our subsequent case studies.

### 2.2 Pro-Active Inference Estimator for Finite-Population Estimands

We propose the following pro-active inference (PAI) estimator after adaptively collecting $n$ labels without replacement:

$$\hat{\theta}_n := \frac{1}{n}\sum_{t=1}^{n}\phi_t, \quad \phi_t := \frac{1}{N}\left(\sum_{i\in O_{t-1}} y_i + \sum_{i\notin O_{t-1}} \hat{f}^{(t-1)}(x_i)\right) + \frac{1}{N}\frac{y_{I_t} - \hat{f}^{(t-1)}(x_{I_t})}{q_t(I_t)}. \tag{2}$$

Intuitively, the first term in $\phi_t$ represents our plug-in estimate for $\theta$—using the labels that we do have and imputing ML predictions for the remaining unobserved labels—while the second

term serves as a bias correction. In Theorem 4, we show that $\hat{\theta}_n$ is unbiased for $\theta$ and derive its limiting distribution via a martingale central limit theorem. Before then, we enumerate some mild regularity conditions below: Assumptions 1-2 are standard conditions for martingale central limit theorems (c.f. Hall and Heyde [2014, Corollary 3.2]) while Assumption 3 is mild and used to facilitate consistency of our variance estimator.

**Assumption 1** (Variance stabilization)**.** *The average conditional variance converges in probability to a positive deterministic limit $\sigma^2$, which does not depend on $m$:*

$$\frac{1}{n}\sum_{t=1}^{n} Var\left(\Delta_t \mid \mathcal{F}_{t-1}\right) \xrightarrow{p} \sigma^2.$$

**Assumption 2** (Lindeberg condition)**.** *The triangular array of problem instances in conjunction with the sampling policies $q_t(\cdot)$ obey the Lindeberg condition. Specifically, for every $\epsilon > 0$,*

$$\frac{1}{n}\sum_{t=1}^{n} \mathbb{E}\left[\Delta_t^2 \mathbf{1}\left(|\Delta_t| > \epsilon\sqrt{n}\right) \mid \mathcal{F}_{t-1}\right] \xrightarrow{p} 0.$$

**Assumption 3** (Uniform fourth-moment control for weighted residuals)**.** *There exists a deterministic constant $C < \infty$ such that*

$$\sup_{m\geq 1}\sup_{1\leq t\leq n_m} \mathbb{E}\left[\left(\frac{1}{N}\frac{y_{I_t} - \hat{f}^{(t-1)}(x_{I_t})}{q_t(I_t)}\right)^4\right] \leq C.$$

**Theorem 4.** *Consider a triangular array of problems (indexed by $m$) such that $n_m \uparrow \infty$, $N_m \uparrow \infty$, and $n_m/N_m \to \rho \in [0, 1-\epsilon]$ for some $\epsilon > 0$.*[1] *For each $t$, suppose that (a) $\hat{f}^{(t-1)}(\cdot)$ and $q_t(\cdot)$ are $\mathcal{F}_{t-1}$-measurable, and that (b) conditional on $\mathcal{F}_{t-1}$, $q_t(\cdot)$ is a probability distribution on $R_{t-1} := \mathcal{I} \setminus O_{t-1}$ with $q_t(i) \geq \tau/|R_{t-1}| = \tau/(N-t+1)$ for all $i \in \mathbb{R}_{t-1}$, for some fixed $\tau \in (0, 1]$. Define $\Delta_t := \phi_t - \theta$.*

*Then, (a) $\hat{\theta}_n$ is unbiased for $\theta$ at every finite $n$; and (b) under mild regularity conditions (variance stabilization, Lindeberg, and uniform fourth-moment control for weighted residuals; Assumptions 1-3), $\hat{\theta}_n$ obeys the following limiting distribution:*

$$\sqrt{n}\left(\hat{\theta}_n - \theta\right)/\hat{\sigma}_n \xrightarrow{d} \mathcal{N}(0, 1), \tag{3}$$

*where letting $\hat{\theta}_{t-1} := \frac{1}{t-1}\sum_{s=1}^{t-1}\phi_s$,*

$$\begin{aligned}\hat{\sigma}_n^2 &:= \hat{A}_n - \hat{B}_n \\ &:= \frac{1}{nN^2}\sum_{t=1}^{n}\frac{\left(y_{I_t} - \hat{f}^{(t-1)}\left(x_{I_t}\right)\right)^2}{q_t(I_t)^2} - \frac{1}{nN^2}\sum_{t=2}^{n}\left(N\hat{\theta}_{t-1} - \sum_{i\in O_{t-1}} y_i - \sum_{i\notin O_{t-1}} \hat{f}^{(t-1)}(x_i)\right)^2.\end{aligned}$$

*Proof.* We begin by showing that the $\Delta_t$ are martingale differences—that their cumulative sums form a martingale. By definition of $\mathcal{F}_{t-1}$ and $q_t(\cdot) := P(I_t = \cdot \mid \mathcal{F}_{t-1})$, with division by $q_t(\cdot)$

[1] We require this last assumption, else sampling without replacement will not having meaningful asymptotics, as we will then collapse towards a census. For readability, we omit the subscripts $m$ henceforth.

well-defined by assumption,

$$\begin{aligned}
\mathbb{E}\left[\Delta_t \mid \mathcal{F}_{t-1}\right] &= \mathbb{E}\left[\frac{1}{N}\left(\sum_{i\in O_{t-1}} y_i + \sum_{i\notin O_{t-1}} \hat{f}^{(t-1)}(x_i)\right) + \frac{1}{N}\frac{y_{I_t} - \hat{f}^{(t-1)}(x_{I_t})}{q_t(I_t)} - \theta \mid \mathcal{F}_{t-1}\right] \\
&= \frac{1}{N}\sum_{i\in O_{t-1}} y_i + \frac{1}{N}\left(\sum_{i\notin O_{t-1}} \hat{f}^{(t-1)}(x_i)\right) + \mathbb{E}\left[\frac{1}{N}\frac{y_{I_t} - \hat{f}^{(t-1)}(x_{I_t})}{q_t(I_t)} \mid \mathcal{F}_{t-1}\right] - \theta \\
&= \frac{1}{N}\sum_{i\in O_{t-1}} y_i + \frac{1}{N}\left(\sum_{i\notin O_{t-1}} \hat{f}^{(t-1)}(x_i)\right) + \frac{1}{N}\sum_{i\notin O_{t-1}}\left(\frac{y_i - \hat{f}^{(t-1)}(x_i)}{q_t(i)}\right) q_t(i) - \theta \\
&= \frac{1}{N}\sum_{i\in\mathcal{I}} y_i - \theta = 0,
\end{aligned}$$

which implies that $(\Delta_t, \mathcal{F}_t)$ is a martingale difference array. Now, it immediately follows that

$$\mathbb{E}\left[\hat{\theta}_n - \theta\right] = \mathbb{E}\left[\frac{1}{n}\sum_{t=1}^{n}\phi_t - \theta\right] = \mathbb{E}\left[\frac{1}{n}\sum_{t=1}^{n}(\phi_t - \theta)\right] = \mathbb{E}\left[\frac{1}{n}\sum_{t=1}^{n}\Delta_t\right] = 0,$$

implying that $\hat{\theta}_n$ is unbiased for $\theta$ at every finite $n$. Now, by Assumptions 1-2, the martingale central limit theorem as presented in Corollary 3.2 of Hall and Heyde [2014] tells us that

$$\sqrt{n}\left(\hat{\theta}_n - \theta\right) \xrightarrow{d} \mathcal{N}\left(0, \sigma^2\right).$$

To compute a consistent estimator for $\sigma^2$, we proceed by computing the martingale increment conditional variances. By definition of $\mathcal{F}_{t-1}$, we have

$$\begin{aligned}
&\mathrm{Var}\left(\Delta_t \mid \mathcal{F}_{t-1}\right) \\
&= \mathrm{Var}\left(\frac{1}{N}\left(\sum_{i\in O_{t-1}} y_i + \sum_{i\notin O_{t-1}} \hat{f}^{(t-1)}(x_i)\right) + \frac{1}{N}\frac{y_{I_t} - \hat{f}^{(t-1)}(x_{I_t})}{q_t(I_t)} - \theta \mid \mathcal{F}_{t-1}\right) \\
&= \mathrm{Var}\left(\frac{1}{N}\frac{y_{I_t} - \hat{f}^{(t-1)}(x_{I_t})}{q_t(I_t)} \mid \mathcal{F}_{t-1}\right) \\
&= \mathbb{E}\left[\left(\frac{1}{N}\frac{y_{I_t} - \hat{f}^{(t-1)}(x_{I_t})}{q_t(I_t)}\right)^2 \mid \mathcal{F}_{t-1}\right] - \left(\mathbb{E}\left[\frac{1}{N}\frac{y_{I_t} - \hat{f}^{(t-1)}(x_{I_t})}{q_t(I_t)} \mid \mathcal{F}_{t-1}\right]\right)^2 \\
&= \frac{1}{N^2}\sum_{i\notin O_{t-1}} \frac{\left(y_i - \hat{f}^{(t-1)}(x_i)\right)^2}{q_t(i)} - \frac{1}{N^2}\left(\sum_{i\notin O_{t-1}}\left(y_i - \hat{f}^{(t-1)}(x_i)\right)\right)^2.
\end{aligned}$$

By Assumption 1 and substitution, we know that

$$\frac{1}{n}\sum_{t=1}^{n}\left(\frac{1}{N^2}\sum_{i\notin O_{t-1}}\frac{\left(y_i-\hat{f}^{(t-1)}(x_i)\right)^2}{q_t(i)}-\frac{1}{N^2}\left(\sum_{i\notin O_{t-1}}\left(y_i-\hat{f}^{(t-1)}(x_i)\right)\right)^2\right)$$

$$=\frac{1}{nN^2}\sum_{t=1}^{n}\left(\sum_{i\notin O_{t-1}}\frac{\left(y_i-\hat{f}^{(t-1)}(x_i)\right)^2}{q_t(i)}\right)-\frac{1}{nN^2}\sum_{t=1}^{n}\left(\sum_{i\notin O_{t-1}}\left(y_i-\hat{f}^{(t-1)}(x_i)\right)\right)^2$$

$$:=A_n-B_n\xrightarrow{p}\sigma^2.$$

To show that $\hat{\sigma}_n^2\xrightarrow{p}\sigma^2$, it suffices to show that (a) $\hat{A}_n-A_n=o_p(1)$ and (b) $\hat{B}_n-B_n=o_p(1)$. Beginning with the first statement,

$$\hat{A}_n-A_n=\frac{1}{nN^2}\sum_{t=1}^{n}\frac{\left(y_{I_t}-\hat{f}^{(t-1)}\left(x_{I_t}\right)\right)^2}{q_t(I_t)^2}-\frac{1}{nN^2}\sum_{t=1}^{n}\left(\sum_{i\notin O_{t-1}}\frac{\left(y_i-\hat{f}^{(t-1)}(x_i)\right)^2}{q_t(i)}\right)$$

$$=\frac{1}{n}\sum_{t=1}^{n}\frac{1}{N^2}\left(\frac{\left(y_{I_t}-\hat{f}^{(t-1)}\left(x_{I_t}\right)\right)^2}{q_t(I_t)^2}-\sum_{i\notin O_{t-1}}\frac{\left(y_i-\hat{f}^{(t-1)}(x_i)\right)^2}{q_t(i)}\right):=\frac{1}{n}\sum_{t=1}^{n}C_t.$$

Because we have

$$\mathbb{E}[C_t\mid\mathcal{F}_{t-1}]=\mathbb{E}\left[\frac{1}{N^2}\left(\frac{\left(y_{I_t}-\hat{f}^{(t-1)}\left(x_{I_t}\right)\right)^2}{q_t(I_t)^2}-\sum_{i\notin O_{t-1}}\frac{\left(y_i-\hat{f}^{(t-1)}(x_i)\right)^2}{q_t(i)}\right)\mid\mathcal{F}_{t-1}\right]$$

$$=\mathbb{E}\left[\frac{1}{N^2}\left(\frac{\left(y_{I_t}-\hat{f}^{(t-1)}\left(x_{I_t}\right)\right)^2}{q_t(I_t)^2}\right)\mid\mathcal{F}_{t-1}\right]-\frac{1}{N^2}\left(\sum_{i\notin O_{t-1}}\frac{\left(y_i-\hat{f}^{(t-1)}(x_i)\right)^2}{q_t(i)}\right)$$

$$=\sum_{i\notin O_{t-1}}\frac{1}{N^2}\left(\frac{\left(y_i-\hat{f}^{(t-1)}\left(x_i\right)\right)^2}{q_t(i)^2}\right)q_t(i)-\frac{1}{N^2}\left(\sum_{i\notin O_{t-1}}\frac{\left(y_i-\hat{f}^{(t-1)}(x_i)\right)^2}{q_t(i)}\right)=0,$$

we can deduce that $(C_t,\mathcal{F}_t)$ is a martingale difference array. By orthogonality of martingale increments (i.e., tower property), we further find that

$$\mathbb{E}\left[\left(\hat{A}_n-A_n\right)^2\right]=\mathbb{E}\left[\left(\frac{1}{n}\sum_{t=1}^{n}C_t\right)^2\right]=\frac{1}{n^2}\sum_{t=1}^{n}\mathbb{E}\left[C_t^2\right].$$

Now, because $(a-b)^2 \leq 2a^2 + 2b^2$,

$$\mathbb{E}\left[C_t^2\right] \leq \mathbb{E}\left[2\left(\frac{1}{N^2}\frac{\left(y_{I_t} - \hat{f}^{(t-1)}\left(x_{I_t}\right)\right)^2}{q_t(I_t)^2}\right)^2 + 2\left(\frac{1}{N^2}\sum_{i\notin O_{t-1}}\frac{\left(y_i - \hat{f}^{(t-1)}(x_i)\right)^2}{q_t(i)}\right)^2\right]$$

$$= 2\mathbb{E}\left[\left(\frac{1}{N}\frac{\left(y_{I_t} - \hat{f}^{(t-1)}\left(x_{I_t}\right)\right)}{q_t(I_t)}\right)^4\right] + 2\mathbb{E}\left[\left(\mathbb{E}\left[\frac{1}{N^2}\frac{\left(y_{I_t} - \hat{f}^{(t-1)}(x_{I_t})\right)^2}{q_t(I_t)^2} \mid \mathcal{F}_{t-1}\right]\right)^2\right]$$

$$= 2\mathbb{E}\left[\left(\frac{1}{N}\frac{\left(y_{I_t} - \hat{f}^{(t-1)}\left(x_{I_t}\right)\right)}{q_t(I_t)}\right)^4\right] + 2\mathbb{E}\left[\mathbb{E}\left[\left(\frac{1}{N}\frac{\left(y_{I_t} - \hat{f}^{(t-1)}(x_{I_t})\right)}{q_t(I_t)}\right)^4 \mid \mathcal{F}_{t-1}\right]\right]$$

$$= 4\mathbb{E}\left[\left(\frac{1}{N}\frac{\left(y_{I_t} - \hat{f}^{(t-1)}\left(x_{I_t}\right)\right)}{q_t(I_t)}\right)^4\right] \leq 4C,$$

with the last step by Assumption 3. As such, we deduce that $\mathbb{E}\left[\left(\hat{A}_n - A_n\right)^2\right] \to 0$, which implies that $\hat{A}_n - A_n = o_p(1)$. Working on the second term, we first observe that

$$B_n = \frac{1}{nN^2}\sum_{t=1}^{n}\left(\sum_{i\notin O_{t-1}}\left(y_i - \hat{f}^{(t-1)}(x_i)\right)\right)^2 = \frac{1}{nN^2}\sum_{t=1}^{n}\left(N\theta - \sum_{i\in O_{t-1}} y_i - \sum_{i\notin O_{t-1}}\hat{f}^{(t-1)}(x_i)\right)^2$$

$$= \frac{1}{nN^2}\sum_{t=2}^{n}\left(N\theta - \sum_{i\in O_{t-1}} y_i - \sum_{i\notin O_{t-1}}\hat{f}^{(t-1)}(x_i)\right)^2 + \frac{1}{nN^2}\left(N\theta - \sum_{i=1}^{N}\hat{f}^{(t-1)}(x_i)\right)^2.$$

With this alternative form, and using the equality that $a^2 - b^2 = (a-b)^2 + 2b(a-b)$ to combine

similar terms, we have the following:

$$\hat{B}_n - B_n = \frac{1}{nN^2}\sum_{t=2}^{n}\left(N\hat{\theta}_{t-1} - \sum_{i\in O_{t-1}} y_i - \sum_{i\notin O_{t-1}} \hat{f}^{(t-1)}(x_i)\right)^2$$

$$-\frac{1}{nN^2}\sum_{t=2}^{n}\left(N\theta - \sum_{i\in O_{t-1}} y_i - \sum_{i\notin O_{t-1}} \hat{f}^{(t-1)}(x_i)\right)^2 - \frac{1}{nN^2}\left(N\theta - \sum_{i=1}^{N}\hat{f}^{(0)}(x_i)\right)^2$$

$$= \frac{1}{nN^2}\sum_{t=2}^{n}\left[\left(N\hat{\theta}_{t-1} - \sum_{i\in O_{t-1}} y_i - \sum_{i\notin O_{t-1}} \hat{f}^{(t-1)}(x_i)\right)^2 - \left(N\theta - \sum_{i\in O_{t-1}} y_i - \sum_{i\notin O_{t-1}} \hat{f}^{(t-1)}(x_i)\right)^2\right]$$

$$-\frac{1}{nN^2}\left(N\theta - \sum_{i=1}^{N}\hat{f}^{(0)}(x_i)\right)^2$$

$$= \frac{1}{n}\sum_{t=2}^{n}\left(\hat{\theta}_{t-1} - \theta\right)^2 + \frac{2}{n}\sum_{t=2}^{n}\left(\theta - \frac{\sum_{i\in O_{t-1}} y_i + \sum_{i\notin O_{t-1}} \hat{f}^{(t-1)}(x_i)}{N}\right)\left(\hat{\theta}_{t-1} - \theta\right)$$

$$-\frac{1}{n}\left(\theta - \frac{\sum_{i=1}^{N}\hat{f}^{(0)}(x_i)}{N}\right)^2.$$

Taking absolute values, we have

$$|\hat{B}_n - B_n| \leq \underbrace{\frac{1}{n}\sum_{t=2}^{n}\left(\hat{\theta}_{t-1} - \theta\right)^2}_{T_1} + \underbrace{\frac{2}{n}\sum_{t=2}^{n}\left|\theta - \frac{\sum_{i\in O_{t-1}} y_i + \sum_{i\notin O_{t-1}} \hat{f}^{(t-1)}(x_i)}{N}\right|\left|\hat{\theta}_{t-1} - \theta\right|}_{T_2}$$

$$+\underbrace{\frac{1}{n}\left(\theta - \frac{\sum_{i=1}^{N}\hat{f}^{(0)}(x_i)}{N}\right)^2}_{T_3}.$$

Working on $T_1$, we have the following because the $\Delta_s$ are martingale differences and by orthogonality of martingale increments:

$$\hat{\theta}_{t-1} - \theta = \frac{1}{t-1}\sum_{s=1}^{t-1}\Delta_s \implies \mathbb{E}\left[\left(\hat{\theta}_{t-1} - \theta\right)^2\right] = \frac{1}{(t-1)^2}\sum_{s=1}^{t-1}\mathbb{E}\left[\Delta_s^2\right].$$

Now, by Jensen's Inequality, the fact that $(a-b)^2 \leq 2a^2 + 2b^2$, and the following fact:

$$\mathbb{E}\left[\frac{1}{N}\frac{y_{I_t} - \hat{f}^{(t-1)}(x_{I_t})}{q_t(I_t)} \mid \mathcal{F}_{t-1}\right] = \frac{1}{N}\sum_{i\notin O_{t-1}}\left(y_i - \hat{f}^{(t-1)}(x_i)\right),$$

we have the following, noting that $\theta - \frac{1}{N}\sum_{i\in O_{s-1}} y_i = \frac{1}{N}\sum_{i\notin O_{s-1}} y_i$:

$$\mathbb{E}\left[\Delta_s^2\right] = \mathbb{E}\left[\left(\frac{1}{N}\left(\sum_{i\in O_{s-1}} y_i + \sum_{i\notin O_{s-1}} \hat{f}^{(s-1)}(x_i)\right) + \frac{1}{N}\frac{y_{I_s} - \hat{f}^{(s-1)}(x_{I_s})}{q_s(I_s)} - \theta\right)^2\right]$$

$$\leq 2\mathbb{E}\left[\left(\frac{1}{N}\frac{y_{I_s} - \hat{f}^{(s-1)}(x_{I_s})}{q_s(I_s)}\right)^2\right] + 2\mathbb{E}\left[\left(\theta - \frac{1}{N}\left(\sum_{i\in O_{s-1}} y_i + \sum_{i\notin O_{s-1}} \hat{f}^{(s-1)}(x_i)\right)\right)^2\right]$$

$$\leq 2\left(\mathbb{E}\left[\left(\frac{1}{N}\frac{y_{I_s} - \hat{f}^{(s-1)}(x_{I_s})}{q_s(I_s)}\right)^4\right]\right)^{1/2} + 2\mathbb{E}\left[\left(\frac{1}{N}\sum_{i\notin O_{s-1}} y_i - \frac{1}{N}\sum_{i\notin O_{s-1}} \hat{f}^{(s-1)}(x_i)\right)^2\right]$$

$$= 2\left(\mathbb{E}\left[\left(\frac{1}{N}\frac{y_{I_s} - \hat{f}^{(s-1)}(x_{I_s})}{q_s(I_s)}\right)^4\right]\right)^{1/2} + 2\mathbb{E}\left[\left(\mathbb{E}\left[\frac{1}{N}\frac{y_{I_s} - \hat{f}^{(s-1)}(x_{I_s})}{q_s(I_s)} \mid \mathcal{F}_{s-1}\right]\right)^2\right]$$

$$\leq 2\left(\mathbb{E}\left[\left(\frac{1}{N}\frac{y_{I_s} - \hat{f}^{(s-1)}(x_{I_s})}{q_s(I_s)}\right)^4\right]\right)^{1/2} + 2\mathbb{E}\left[\mathbb{E}\left[\left(\frac{1}{N}\frac{y_{I_s} - \hat{f}^{(s-1)}(x_{I_s})}{q_s(I_s)}\right)^2 \mid \mathcal{F}_{s-1}\right]\right]$$

$$\leq 2\left(\mathbb{E}\left[\left(\frac{1}{N}\frac{y_{I_s} - \hat{f}^{(s-1)}(x_{I_s})}{q_s(I_s)}\right)^4\right]\right)^{1/2} + 2\mathbb{E}\left[\left(\frac{1}{N}\frac{y_{I_s} - \hat{f}^{(s-1)}(x_{I_s})}{q_s(I_s)}\right)^2\right]$$

$$\leq 2\left(\mathbb{E}\left[\left(\frac{1}{N}\frac{y_{I_s} - \hat{f}^{(s-1)}(x_{I_s})}{q_s(I_s)}\right)^4\right]\right)^{1/2} + 2\left(\mathbb{E}\left[\left(\frac{1}{N}\frac{y_{I_s} - \hat{f}^{(s-1)}(x_{I_s})}{q_s(I_s)}\right)^4\right]\right)^{1/2} \leq 4C^{1/2}.$$

Thus, we conclude that

$$\mathbb{E}[T_1] = \frac{1}{n}\sum_{t=2}^{n}\mathbb{E}\left[\left(\hat{\theta}_{t-1} - \theta\right)^2\right] = \frac{1}{n}\sum_{t=2}^{n}\frac{1}{(t-1)^2}\sum_{s=1}^{t-1}\mathbb{E}\left[\Delta_s^2\right] \leq \frac{1}{n}\sum_{t=2}^{n}\frac{4C^{1/2}}{t-1} = O\left(\frac{\log n}{n}\right).$$

Moving on to $T_3$, we note that for any $t$, by definition of $\theta$,

$$\theta - \frac{1}{N}\sum_{i\in O_{t-1}} y_i - \frac{1}{N}\sum_{i\notin O_{t-1}} \hat{f}^{(t-1)}(x_i) = \frac{1}{N}\sum_{i\notin O_{t-1}}\left(y_i - \hat{f}^{(t-1)}(x_i)\right)$$

$$= \mathbb{E}\left[\frac{1}{N}\frac{y_{I_t} - \hat{f}^{(t-1)}(x_{I_t})}{q_t(I_t)} \mid \mathcal{F}_{t-1}\right].$$

By Jensen's inequality, we deduce that for any $t$,

$$\mathbb{E}\left[\left(\theta - \frac{1}{N}\sum_{i\in O_{t-1}} y_i - \frac{1}{N}\sum_{i\notin O_{t-1}} \hat{f}^{(t-1)}(x_i)\right)^2\right] = \mathbb{E}\left[\left(\mathbb{E}\left[\frac{1}{N}\frac{y_{I_t} - \hat{f}^{(t-1)}(x_{I_t})}{q_t(I_t)} \mid \mathcal{F}_{t-1}\right]\right)^2\right]$$

$$\leq \mathbb{E}\left[\mathbb{E}\left[\left(\frac{1}{N}\frac{y_{I_t} - \hat{f}^{(t-1)}(x_{I_t})}{q_t(I_t)}\right)^2 \mid \mathcal{F}_{t-1}\right]\right] = \mathbb{E}\left[\left(\frac{1}{N}\frac{y_{I_t} - \hat{f}^{(t-1)}(x_{I_t})}{q_t(I_t)}\right)^2\right]$$

$$\leq \left(\mathbb{E}\left[\left(\frac{1}{N}\frac{y_{I_t} - \hat{f}^{(t-1)}(x_{I_t})}{q_t(I_t)}\right)^4\right]\right)^{1/2} \leq C^{1/2}.$$

Thus, it also follows that, corresponding to $t = 1$,

$$\mathbb{E}[T_3] = \mathbb{E}\left[\frac{1}{n}\left(\theta - \frac{\sum_{i=1}^{N} \hat{f}^{(0)}(x_i)}{N}\right)^2\right] \leq \frac{C^{1/2}}{n} = O\left(\frac{1}{n}\right).$$

Now, working on $T_2$, by Cauchy-Schwarz, and using our above facts,

$$\begin{aligned}
\mathbb{E}[T_2] &= \frac{2}{n}\sum_{t=2}^{n} \mathbb{E}\left[\left|\theta - \frac{\sum_{i \in O_{t-1}} y_i + \sum_{i \notin O_{t-1}} \hat{f}^{(t-1)}(x_i)}{N}\right| \left|\hat{\theta}_{t-1} - \theta\right|\right] \\
&\leq \frac{2}{n}\sum_{t=2}^{n} \mathbb{E}\left[\left(\theta - \frac{\sum_{i \in O_{t-1}} y_i + \sum_{i \notin O_{t-1}} \hat{f}^{(t-1)}(x_i)}{N}\right)^2\right]^{1/2} \mathbb{E}\left[\left(\hat{\theta}_{t-1} - \theta\right)^2\right]^{1/2} \\
&\leq \frac{2}{n}\sum_{t=2}^{n} \left(C^{1/2}\right)^{1/2} \left(\frac{4C^{1/2}}{t-1}\right)^{1/2} = \frac{4C^{1/2}}{n}\sum_{t=2}^{n} \frac{1}{\sqrt{t-1}} = O\left(\frac{1}{\sqrt{n}}\right).
\end{aligned}$$

Putting everything together, we deduce that

$$\mathbb{E}[|\hat{B}_n - B_n|] \leq \mathbb{E}[T_1] + \mathbb{E}[T_2] + \mathbb{E}[T_3] = O\left(\frac{\log n}{n}\right) + O\left(\frac{1}{\sqrt{n}}\right) + O\left(\frac{1}{n}\right) \to 0,$$

which implies that $\hat{B}_n \xrightarrow{L^1} B_n$, which implies $\hat{B}_n - B_n = o_p(1)$. Thus, we conclude that $\hat{\sigma}_n^2 \xrightarrow{p} \sigma^2$, which by Slutsky's gives us our desired final result:

$$\sqrt{n}\left(\hat{\theta}_n - \theta\right)/\hat{\sigma}_n \xrightarrow{d} \mathcal{N}(0,1).$$

□

### 2.3 Pro-Active Inference Estimator for Superpopulation Estimands

## 3 Predicting Poverty Rates

## 4 Predicting Educational Outcomes

## References

P. Hall and C. C. Heyde. *Martingale Limit Theory and its Application*. Academic Press, 2014.